

Christof Teuscher

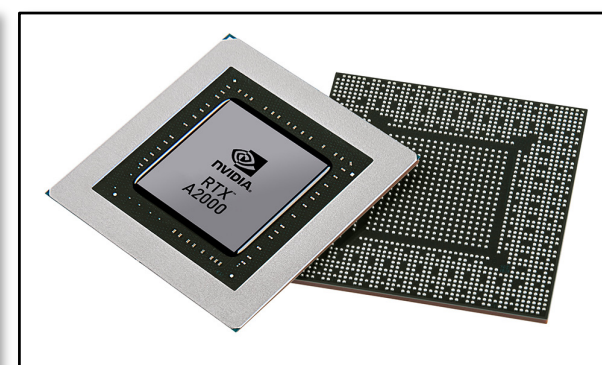
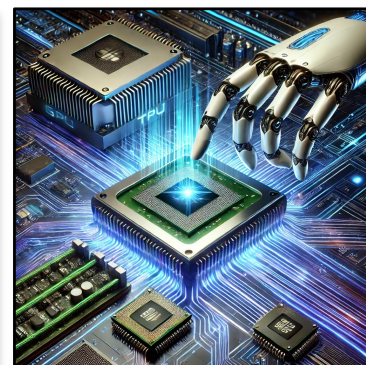
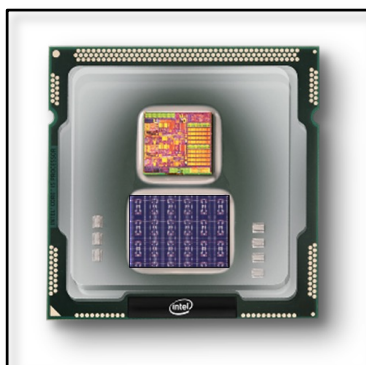
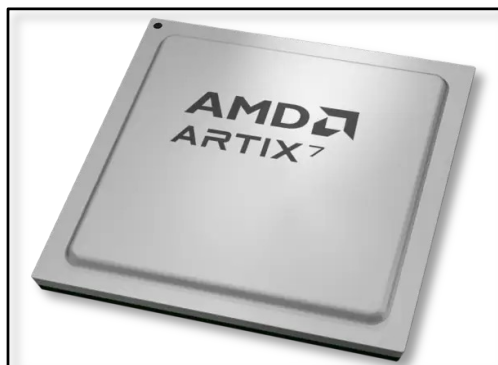
ECE 410/510: Hardware for AI and ML, Spring 2026

Welcome and Introduction

Portland State University
Department of Electrical and Computer Engineering (ECE)

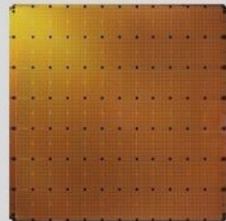
www.teuscher-lab.com

teuscher@pdx.edu



The most complex human-made systems ever created

CS-3: Twice the speed at same power & cost



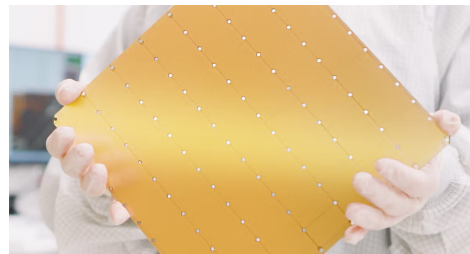
**Cerebras
Wafer Scale Engine 3**

WSE-3

- 4 trillion transistors
- 46,225 mm² silicon
- 900,000 cores optimized for sparse linear algebra
- 125 petaFLOPS of AI compute
- 44 GB of on-chip memory
- 21 PB/s memory bandwidth
- 214 Pb/s fabric bandwidth
- TSMC 5nm process

Wafer Scale Engine-3 vs. H100

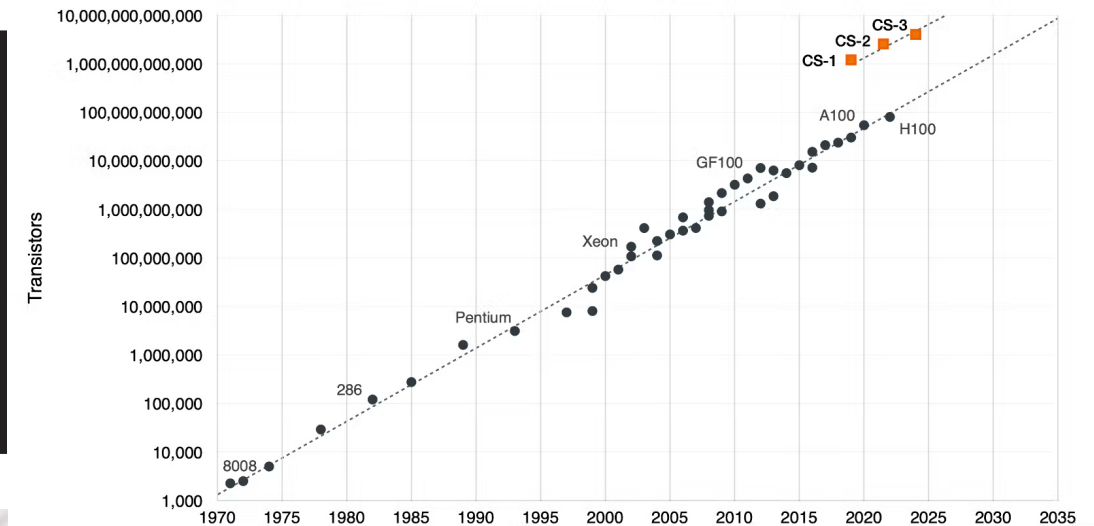
	Cerebras WSE-3	Nvidia H100	Cerebras Advantage
Chip size	46,225 mm ²	814 mm ²	57x
Cores	900,000	16,896 FP32 + 528 Tensor	52x
On-chip memory	44 Gigabytes	0.05 Gigabytes	880x
Memory bandwidth	21 Petabytes/sec	0.003 Petabytes/sec	7,000X
Fabric bandwidth	214 Petabits/sec	0.0576 Petabits/sec	3,715X



The human brain consists of 100 billion neurons and over 100 trillion synaptic connections.

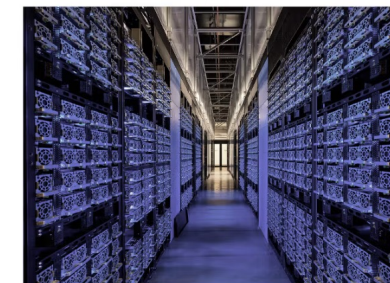
<https://www.cerebras.ai/blog/cerebras-cs3>

Cerebras Wafer Scale Engine – The Chip That Broke Moore's Law



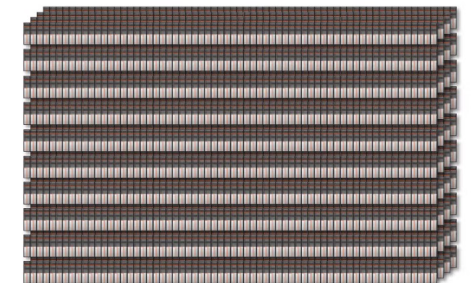
Train Llama2-70B in a Single Day

Meta GPU Cluster



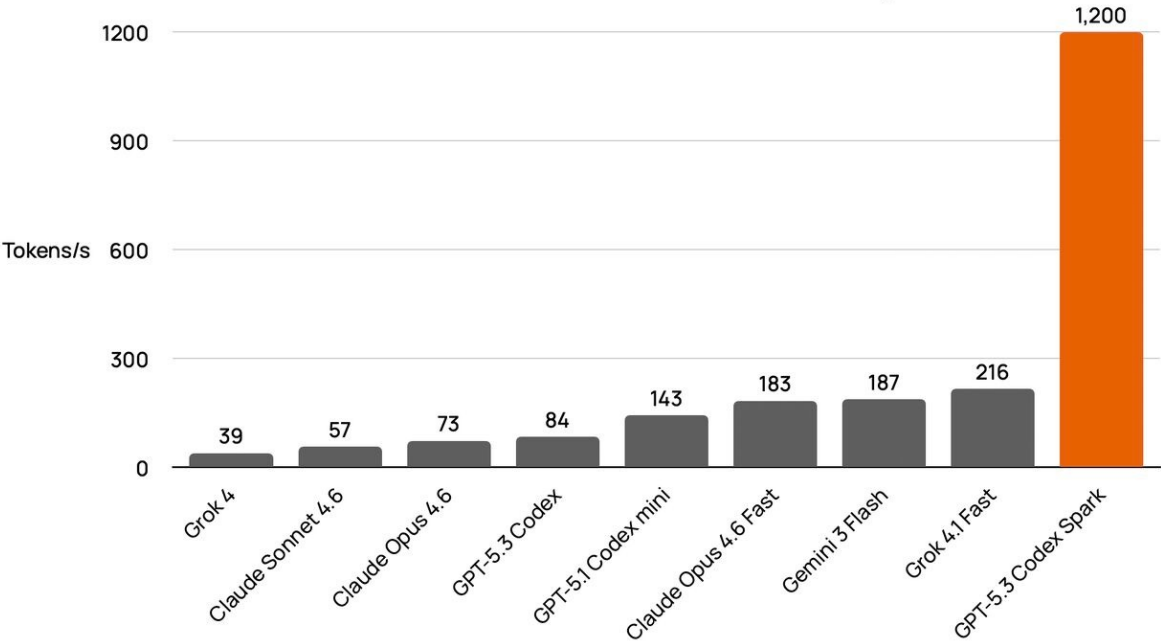
~1 month

Cerebras CS-3 Cluster



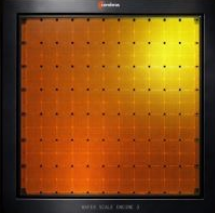
~1 day

Cerebras Powers the World's Fastest Coding Model



Cerebras is coming to AWS





AWS TRAINIUM 3

CEREBRAS WSE 3

One of NVIDIA's big GTC announcements was a \$20 billion bet on the same problem we solved 6 years ago.

Their next-gen inference chip – not available yet – has 140x less memory bandwidth than **Cerebras** has today.

To run a single 2 trillion parameter model, you need more than 2,000 Groq chips. On Cerebras, only 20 wafers.

Even paired with GPUs, in a solution that is scheduled to be delivered later this year, the solution maxes out at ~1,000 tokens per second.

We run at thousands of tokens per second today. And every day. In production now.

When you connect 2,000 little chips together, every interconnect adds latency. Every cable has overhead. It doesn't matter what your memory bandwidth is on paper if you're bottlenecked by the wiring between thousands of tiny chips.

We solved this with wafer scale. One integrated system. Little interconnect tax.

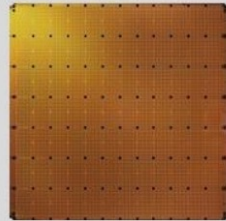
NVIDIA told the world that fast inference is where the value is.

They are right – it's why leading AI companies and hyperscalers are choosing **Cerebras**.

Cerebras vs. Rubin + Groq

	Cerebras CS-3	Rubin + Groq 3 LPU	Cerebras Advantage
Memory	44GB	0.5GB	90X
Chips Needed for 2T Parameter Model	23	2000	90X
Memory Bandwidth	21 Petabytes/s	150 Terabytes/s	140X
Availability	Now	End 2026	

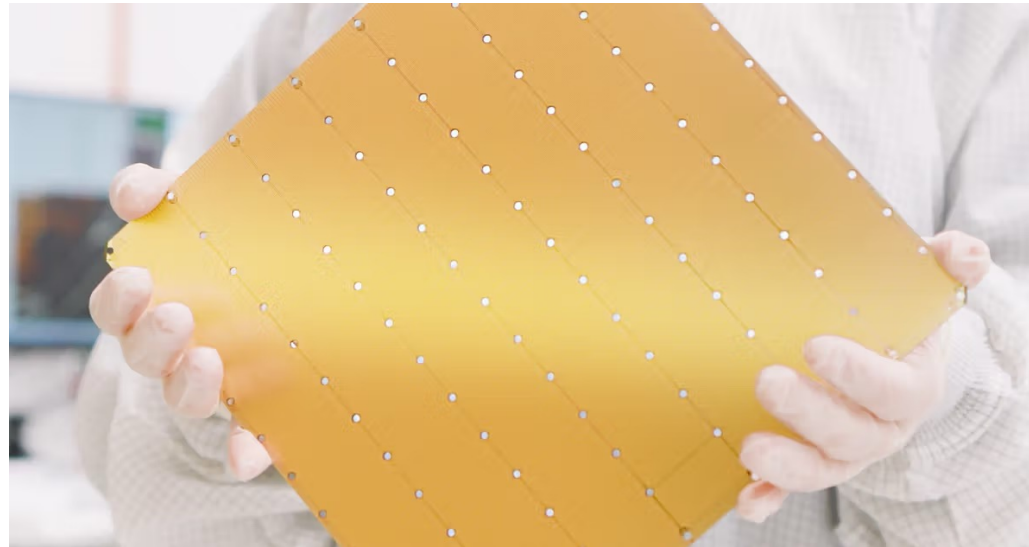
How much power does this chip use?



**Cerebras
Wafer Scale Engine 3**

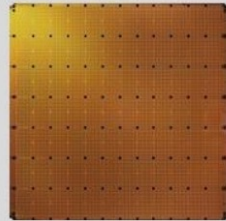
WSE-3

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<https://www.wsj.com/opinion/the-microchip-era-is-about-to-end-e71eb66a>

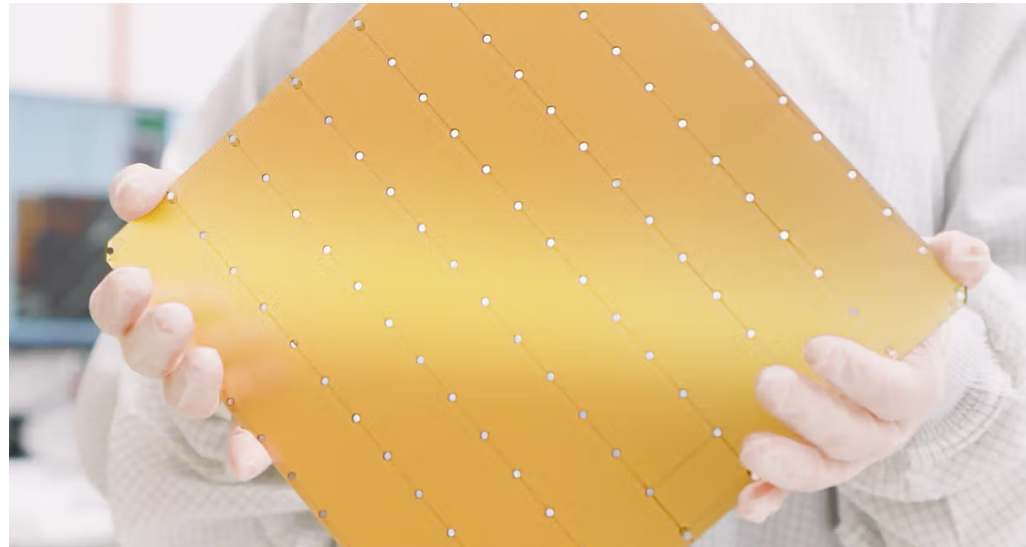
How much power does this chip use?



Cerebras
Wafer Scale Engine 3

WSE-3

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<https://www.wsj.com/opinion/the-microchip-era-is-about-to-end-e71eb66a>

- 23 kW
- Closed internal water cooling
- A typical U.S. house experiences peak power consumption between 5 and 10kW.

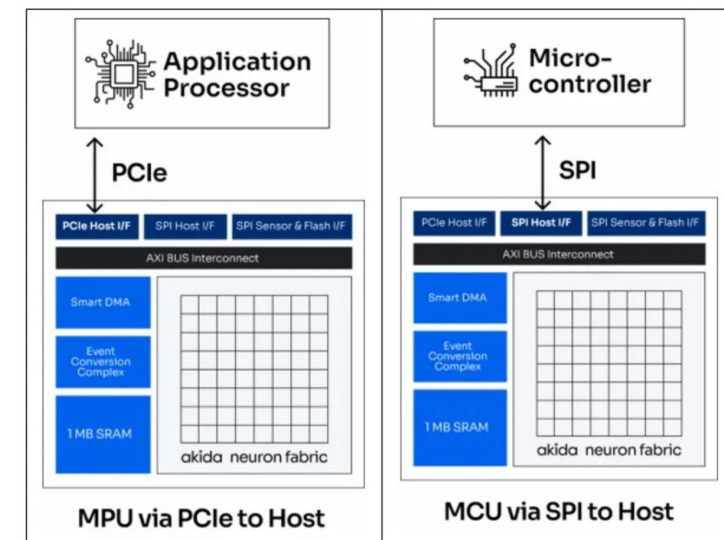
BrainChip launches AKD1500 Edge AI co-processor

BrainChip launches AKD1500 Edge AI co-processor

Artificial Intelligence 18 March 2026 by News Desk



The AKD1500 achieves a breakthrough in efficiency for low power Edge applications by delivering slightly under **1 tera operations per second (TOPS)** while consuming less than **200mW** of power in serial mode and 300mW in PCIe mode.



<https://www.electronicsspecifier.com/products/artificial-intelligence/brainchip-launches-akd1500-edge-ai-co-processor>

The New York Times

U.S. Economy | Inflation | Energy Costs | Interest Rates | Mortgage Rates | Jobs Losses

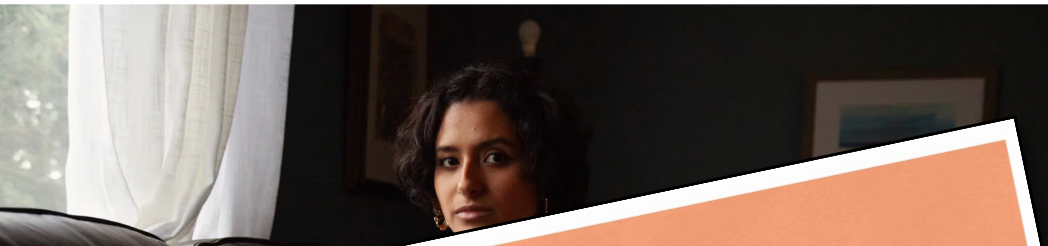
Young Graduates Face the Grimmiest Job Market in Years

Artificial intelligence could reshape work, but for now a low-hire, low-fire labor market is the main impediment for young people seeking employment.

Listen · 8:50 min

Share full article

1.2K



Why College Graduates Feel Betrayed

Their anger goes far beyond the recent rise of unemployment and the looming threat of A.I.

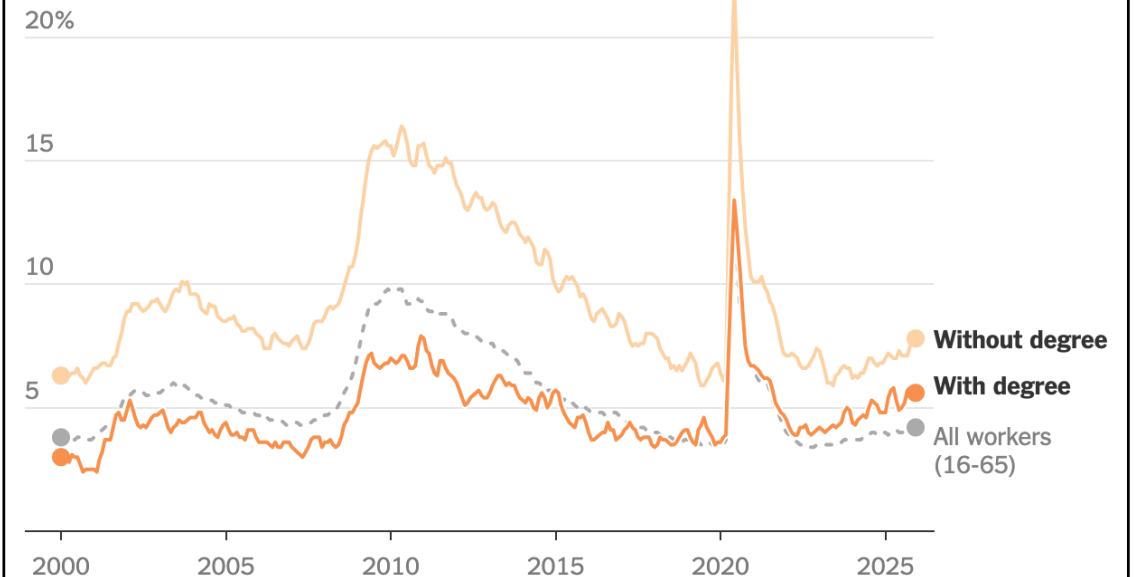
10 MIN READ



Marcos Chin

Unemployment rates for young people

Among workers with and without college degrees, between the ages of 22 and 27



Notes: Data is seasonally adjusted and based on a three-month moving average. Source: Federal Reserve Bank of New York. The New York Times

<https://www.nytimes.com/2026/03/24/business/economy/college-graduates-job-market-hiring.html>

<https://www.nytimes.com/2026/03/27/business/college-graduates-economy-unemployment-.html>

AI proficiency is not optional anymore!

THE WALL STREET JOURNAL.

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ECONOMY | JOBS | CAPITAL ACCOUNT

Tech Has Never Caused a Job Apocalypse. Don't Bet on It Now.

Neither theory, history nor the latest data suggests a recession driven by AI job dislocation is likely



By Greg Ip [Follow](#)

Feb. 27, 2026 5:30 am ET



Share



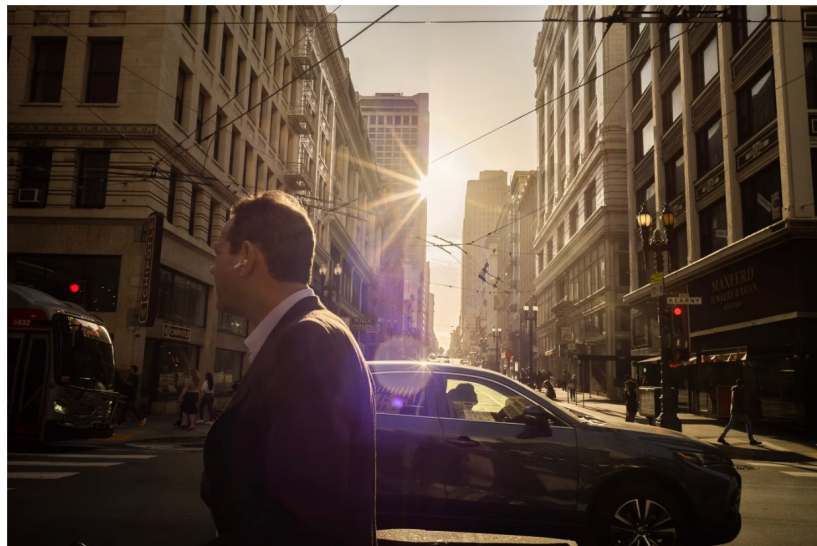
Resize



589

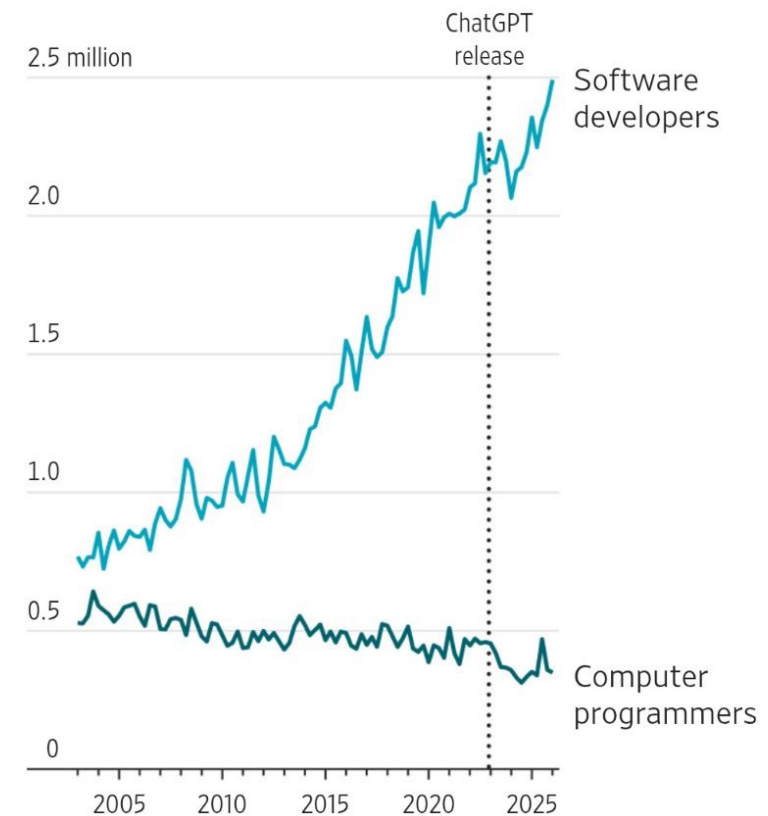


Listen (1 min)



People should still have jobs to commute to despite AI. BRIAN L. FRANK FOR WSJ

U.S. employment, quarterly



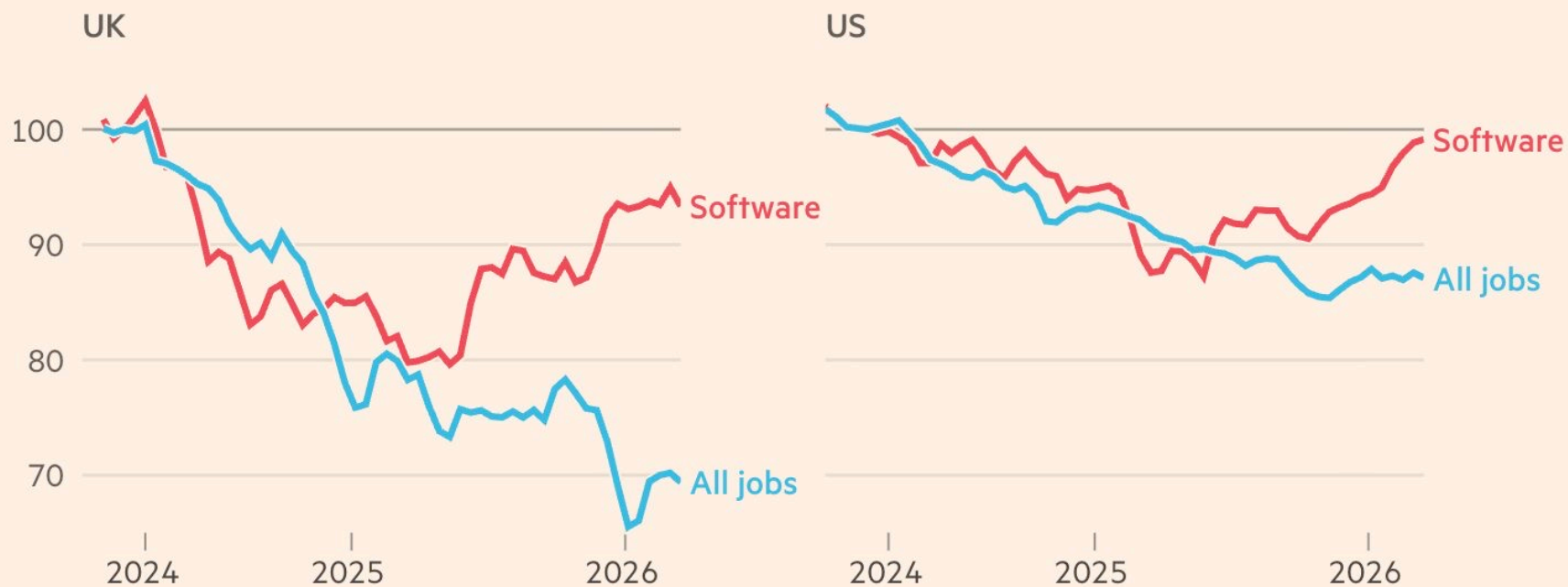
Note: Quarterly averages; 1Q 2026 figure is as of January. Not seasonally adjusted.

Source: Labor Department via James Bessen, Technology and Policy Research Initiative at Boston University

<https://www.wsj.com/economy/jobs/tech-has-never-caused-a-job-apocalypse-dont-bet-on-it-now-d192b579>

Software jobs are rebounding, bucking the overall trend of weak hiring

Job postings volume (Mar 2024 = 100)



Source: FT analysis of Indeed Hiring Lab data
FT graphic: John Burn-Murdoch / @jburnmurdoch
©FT

<https://www.ft.com/content/7325e967-5f4e-40b1-af3f-7d2351781843?syn-25a6b1a6=1>



DESIGNLINES | AI & BIG DATA DESIGNLINE

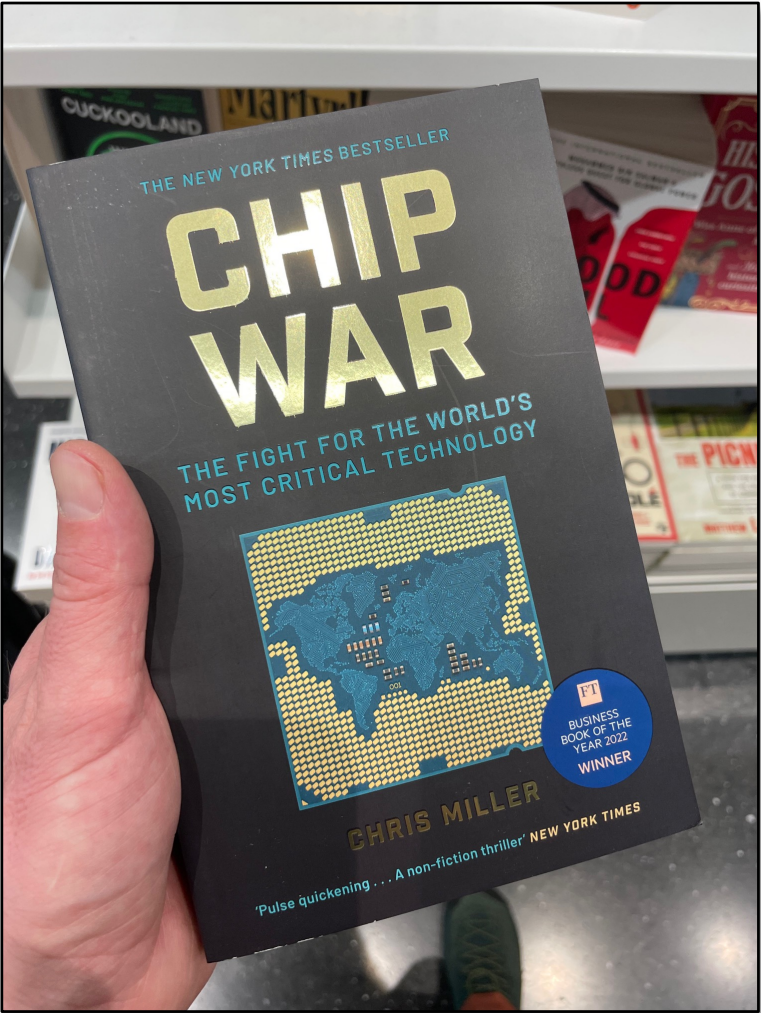
TSMC Price Hikes End the Era of Cheap Transistors

By Pablo Valerio 10.01.2025 0

Process Node	Year of HVM Introduction	Estimated Avg. Wafer Price (USD)	Node-over-Node Price Increase (%)	Key Customers at Launch
7nm	2018	~\$9,350	-	Apple, AMD, Nvidia
5nm	2020	~\$17,000	~82%	Apple, AMD
3nm	2022-2023	~\$20,000	~18%	Apple
2nm	2025 (est.)	~\$30,000+	~50%+	Apple, Nvidia, AMD, MediaTek

Semiconductor Wafer Cost Evolution by Process Node (2020-2026E)

<https://www.eetimes.com/tsmc-price-hikes-end-the-era-of-cheap-transistors>



Geopolitical dimensions of the fight for AI dominance

Commodore C64



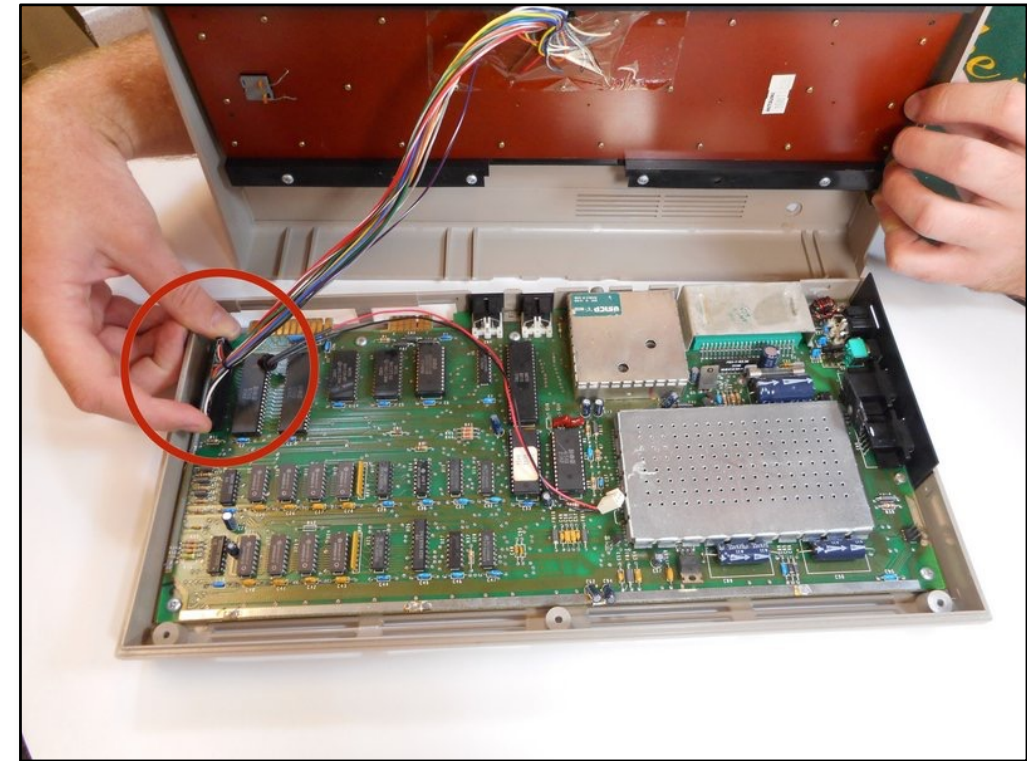
1982



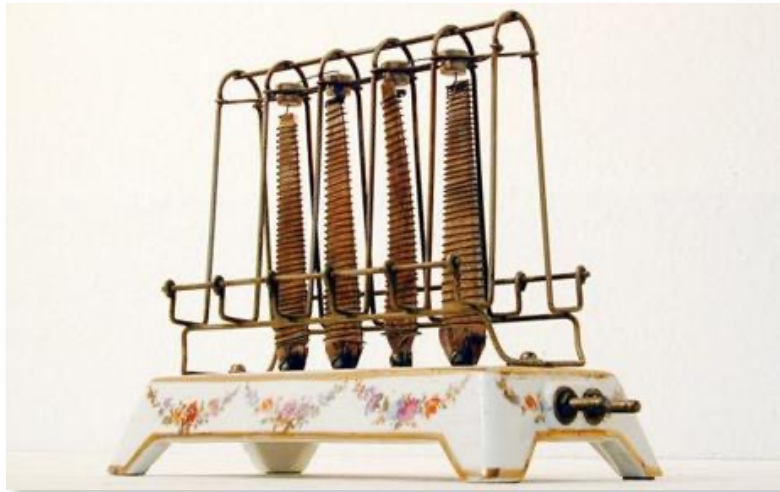
Radio-frequency
identification (RFID) tag

Same amount of memory!

The way we build, program, and use computers has fundamentally changed



MOS 6510, 1 MHz, 64 kB RAM, ~9000 transistors





Daily Mail.com

The perfect piece of toast: Scientists test 2,000 slices and find 216 seconds is the optimum time

By DAILY MAIL REPORTER
UPDATED: 04:03 EST, 22 July 2011



136
View comments

Scientists today revealed the mathematical formula for a perfect slice of toast, showing that it is best cooked for exactly 216 seconds.

A team of researchers carried out a study which found the optimum thickness is 14mm and the ideal amount of butter is 0.44 grams per square inch.

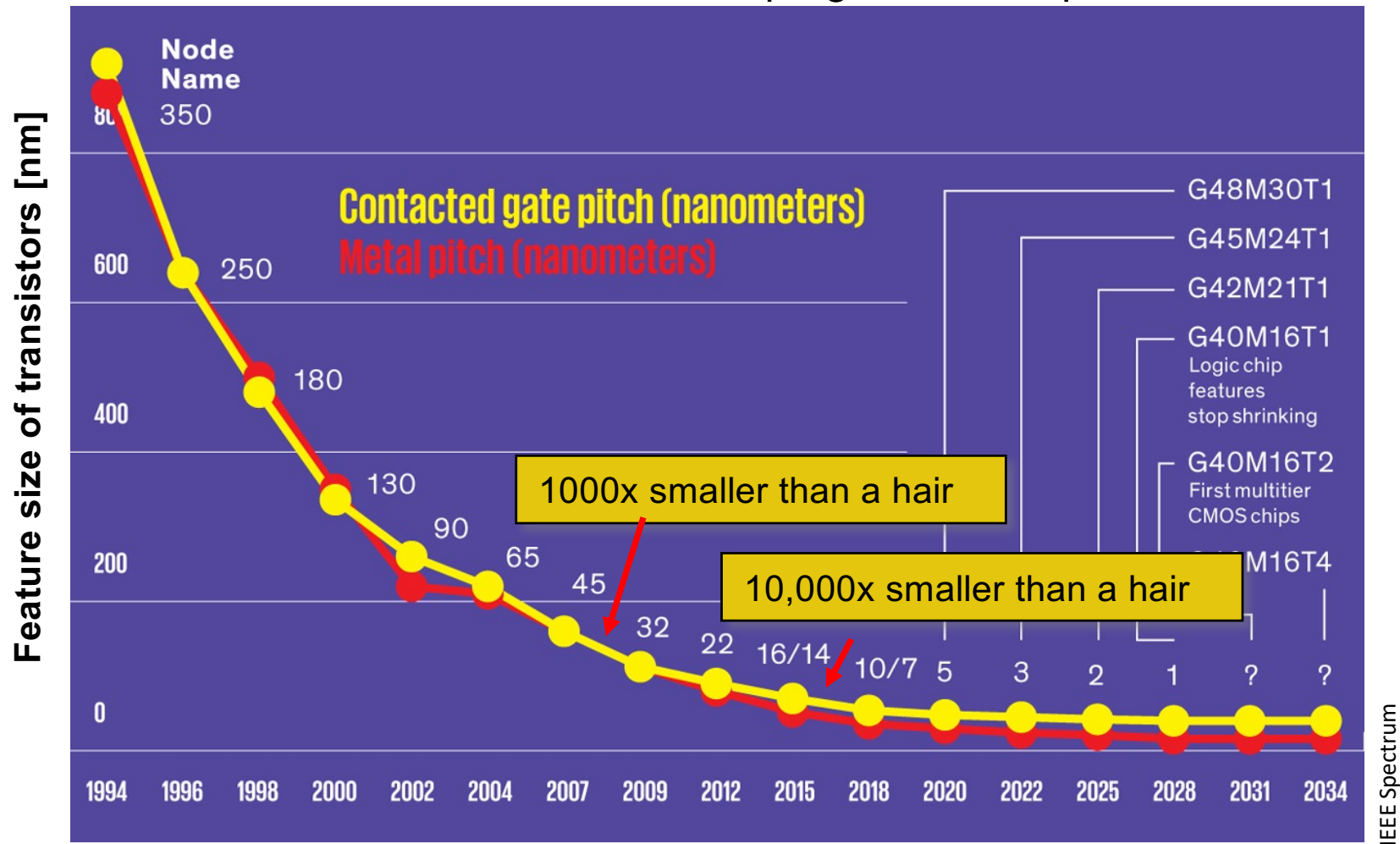
The recommended cooking time gives the slice a 'golden-brown' colour and the 'ultimate balance of external crunch and internal softness'.



Perfect: The ideal slice of toast is cooked for just over three minutes, according to scientists

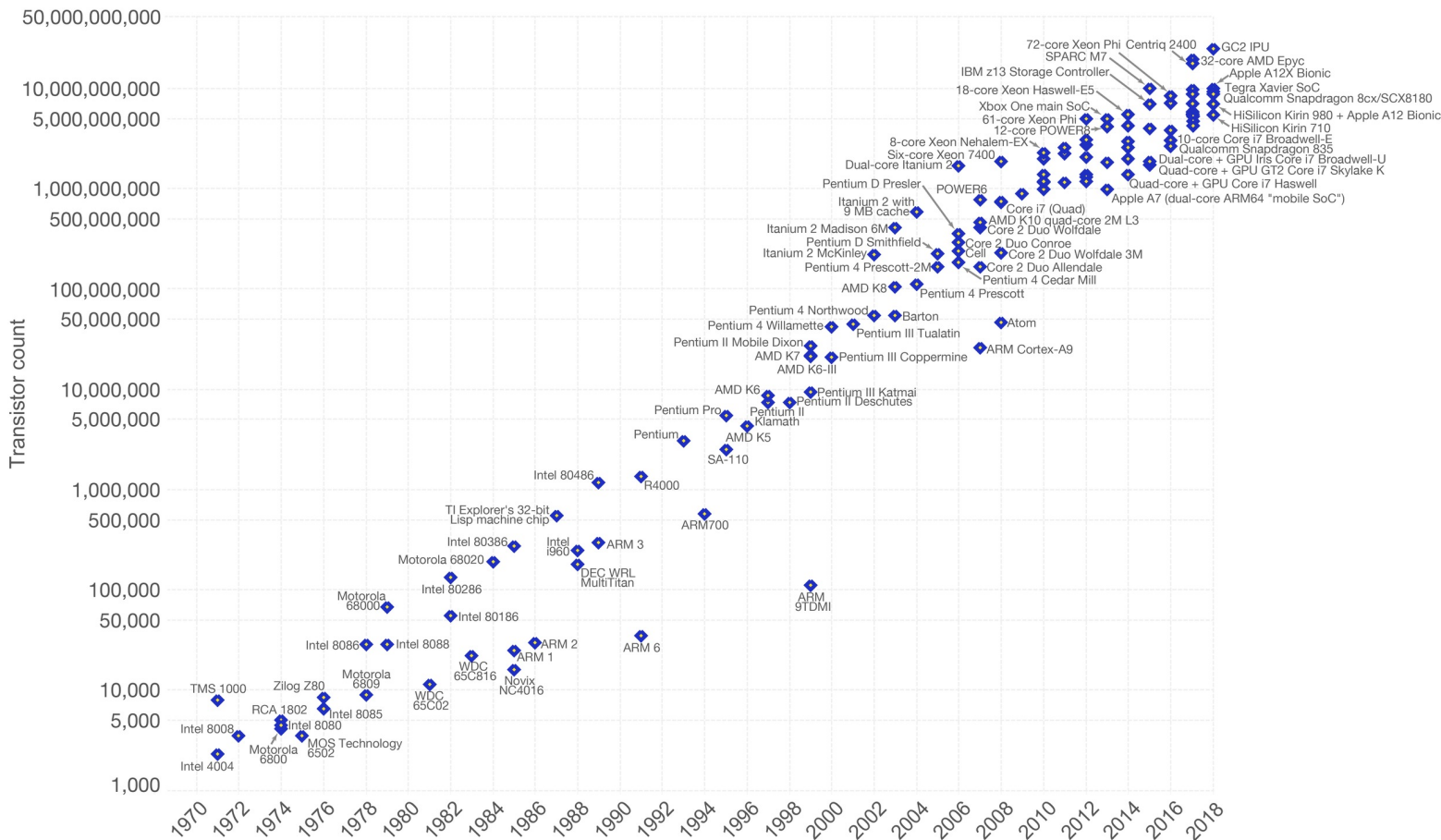
Smaller smaller smaller smaller smaller smaller smaller smaller smaller

Miniaturization: the cornerstone of progress in computer science



Moore’s Law – The number of transistors on integrated circuit chips (1971-2018)

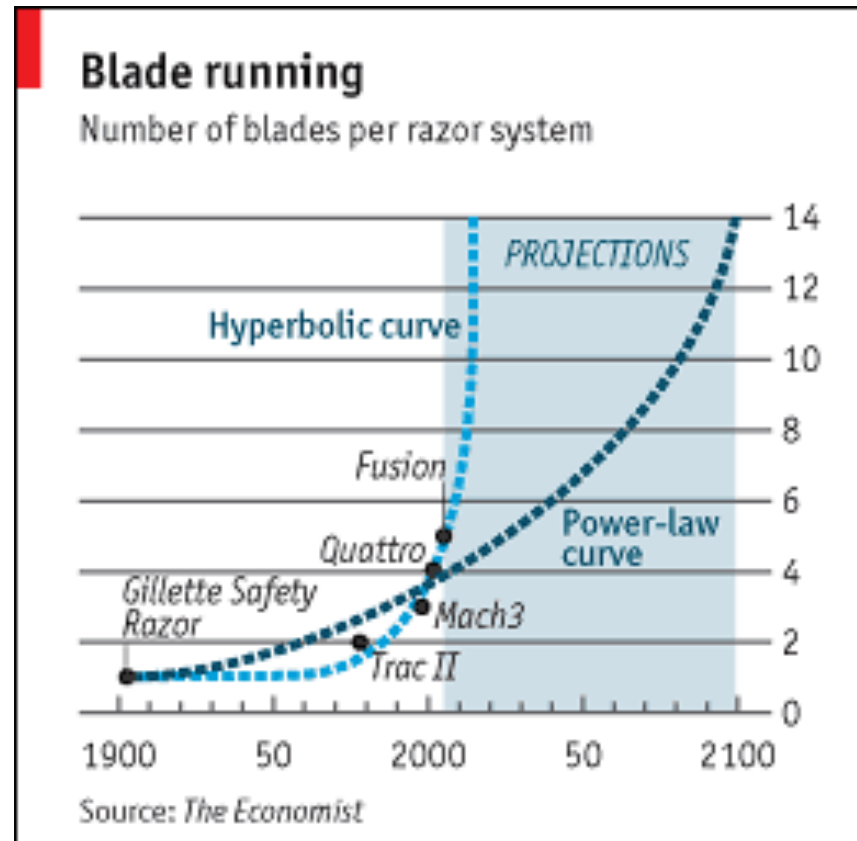
Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important as other aspects of technological progress – such as processing speed or the price of electronic products – are linked to Moore's law.



Data source: Wikipedia (https://en.wikipedia.org/wiki/Transistor_count)
The data visualization is available at [OurWorldinData.org](https://www.ourworldindata.org). There you find more visualizations and research on this topic.

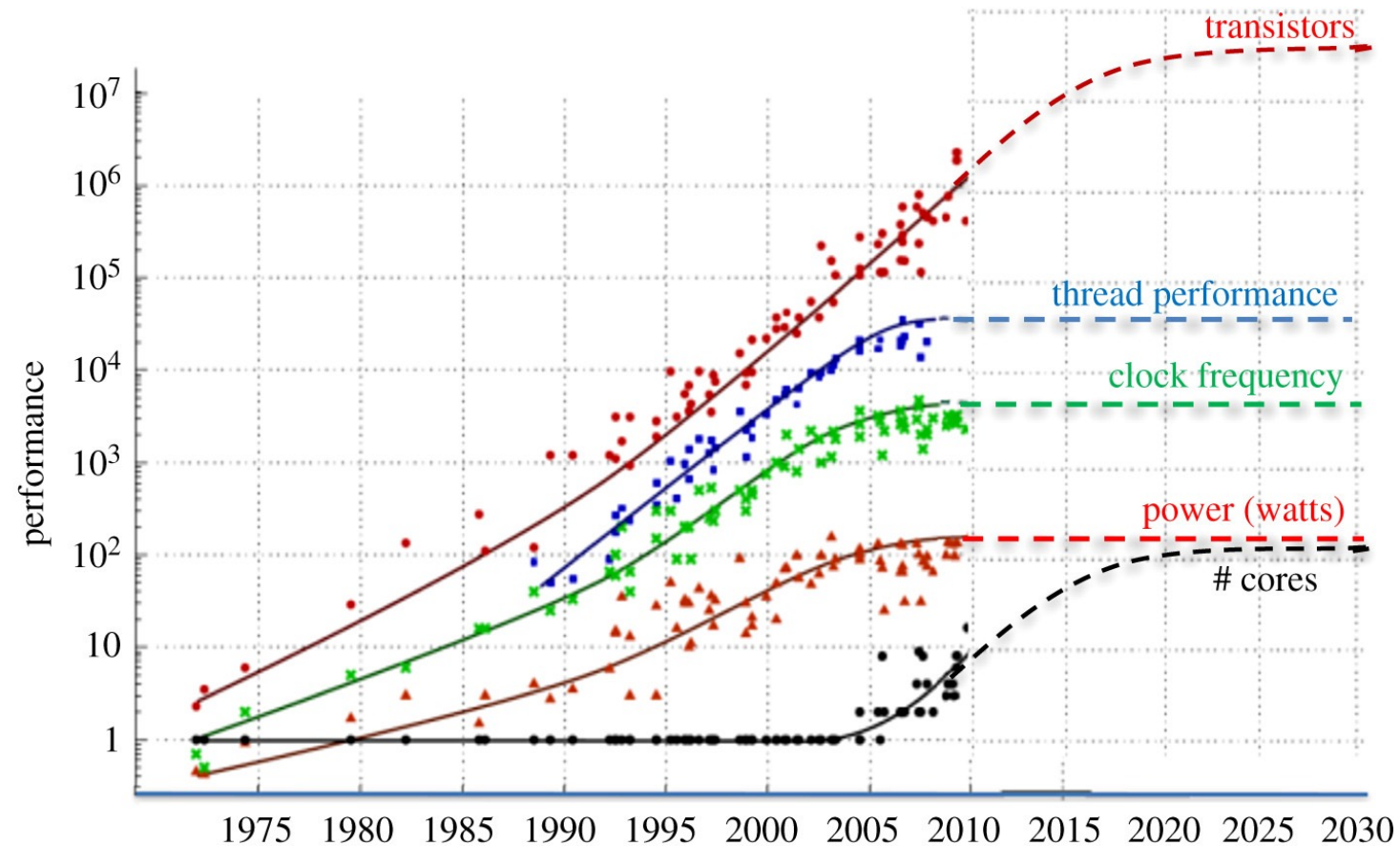
Licensed under CC-BY-SA by the author Max Roser.

The Cutting Edge: A Moore's Law for Razor Blades?

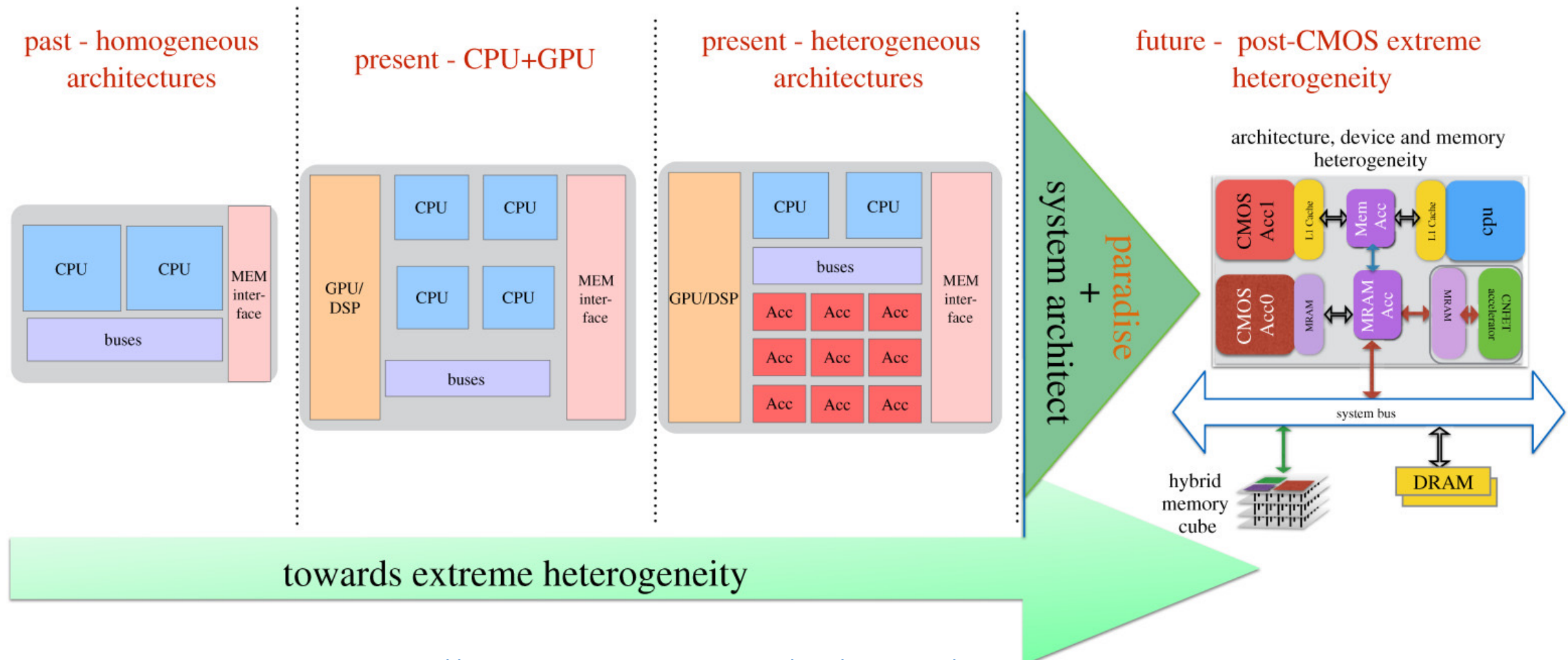


Source: *The Economist*,
March 16, 2006

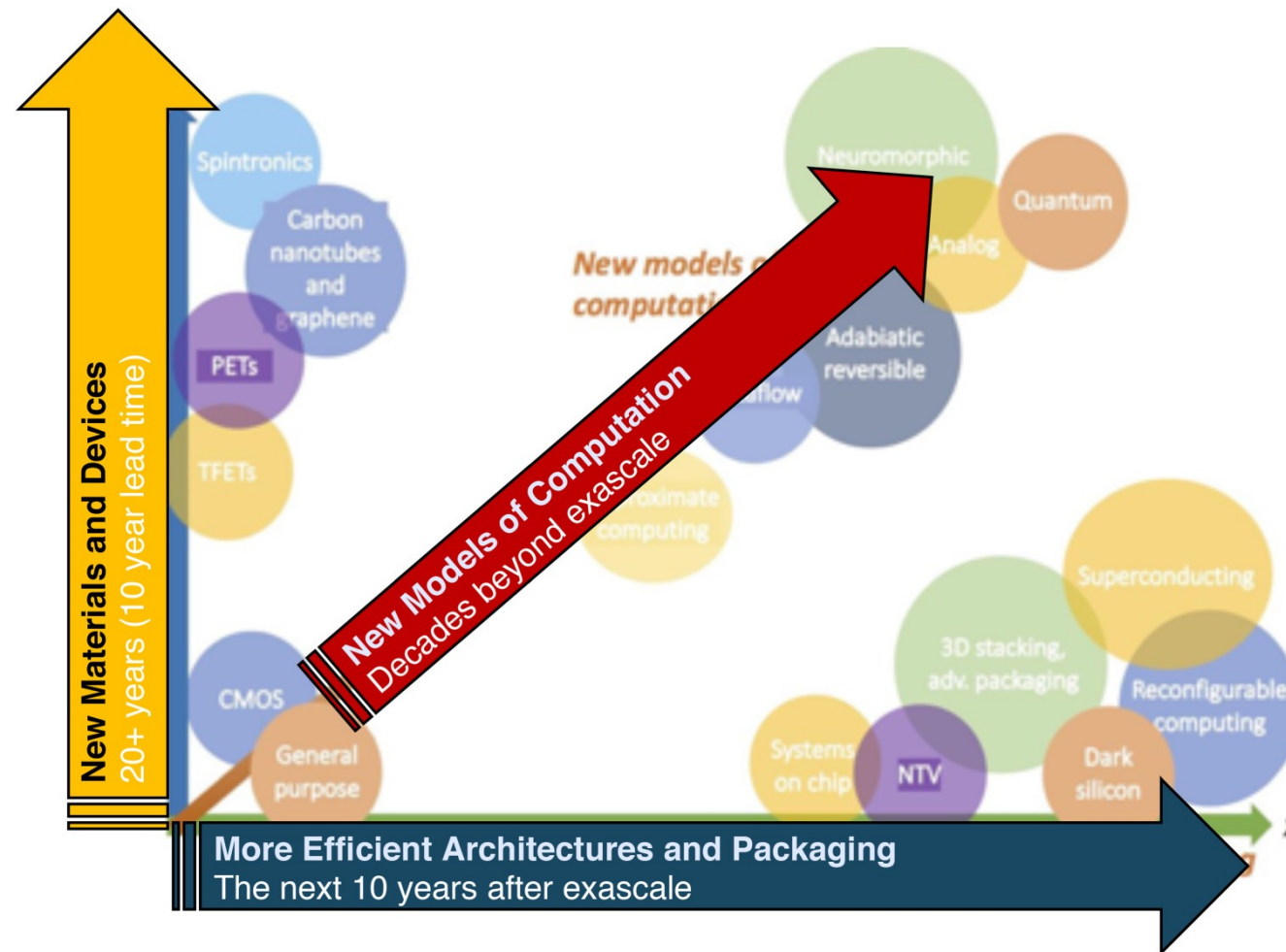
More of Moore?



<https://royalsocietypublishing.org/doi/10.1098/rsta.2019.0061>



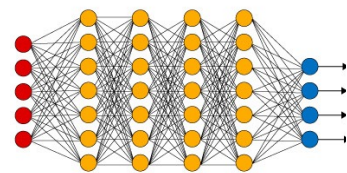
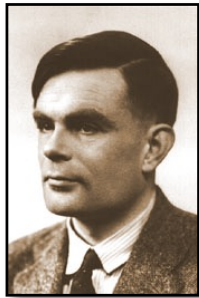
<https://royalsocietypublishing.org/doi/10.1098/rsta.2019.0061>



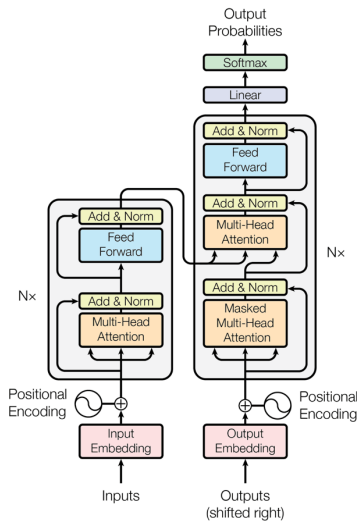
<https://royalsocietypublishing.org/doi/10.1098/rsta.2019.0061>

How did we get here?

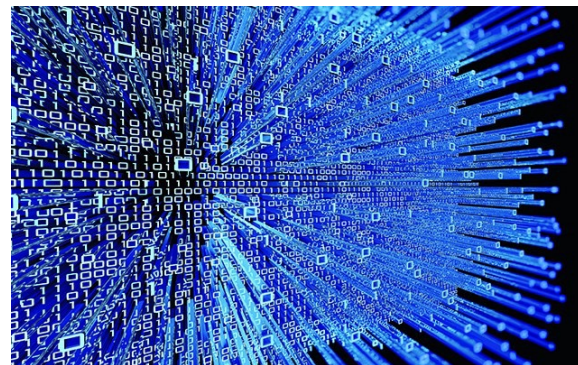
Alan Turing



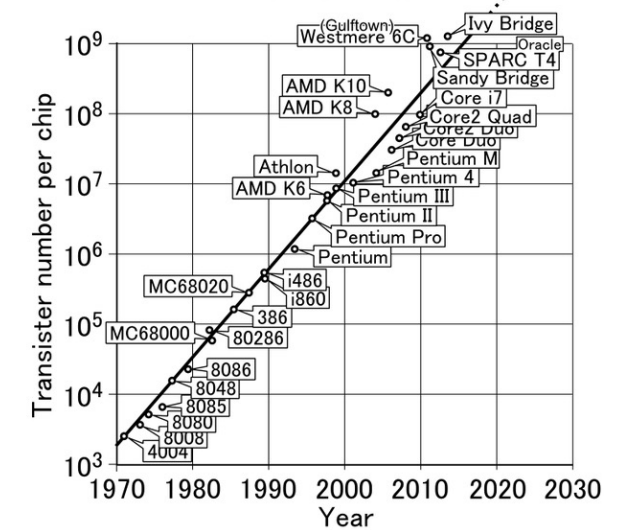
Deep neural networks



Transformers



(Big) data



Moore's law



Where is this going?

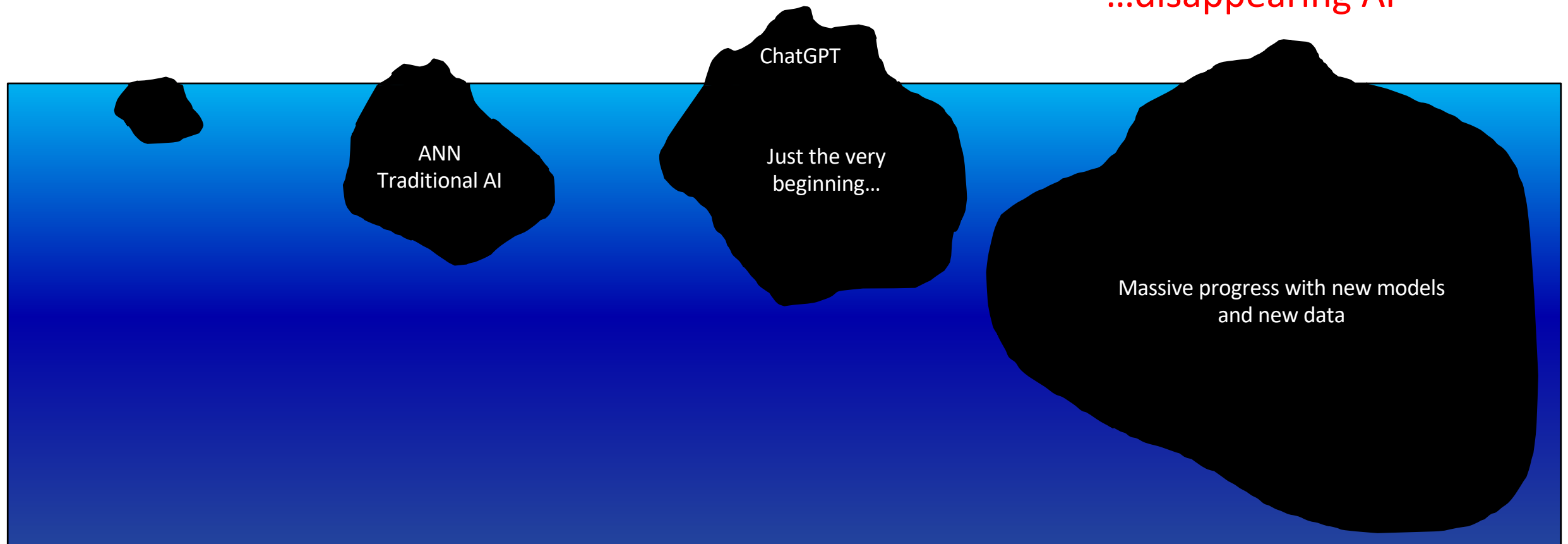
50ies

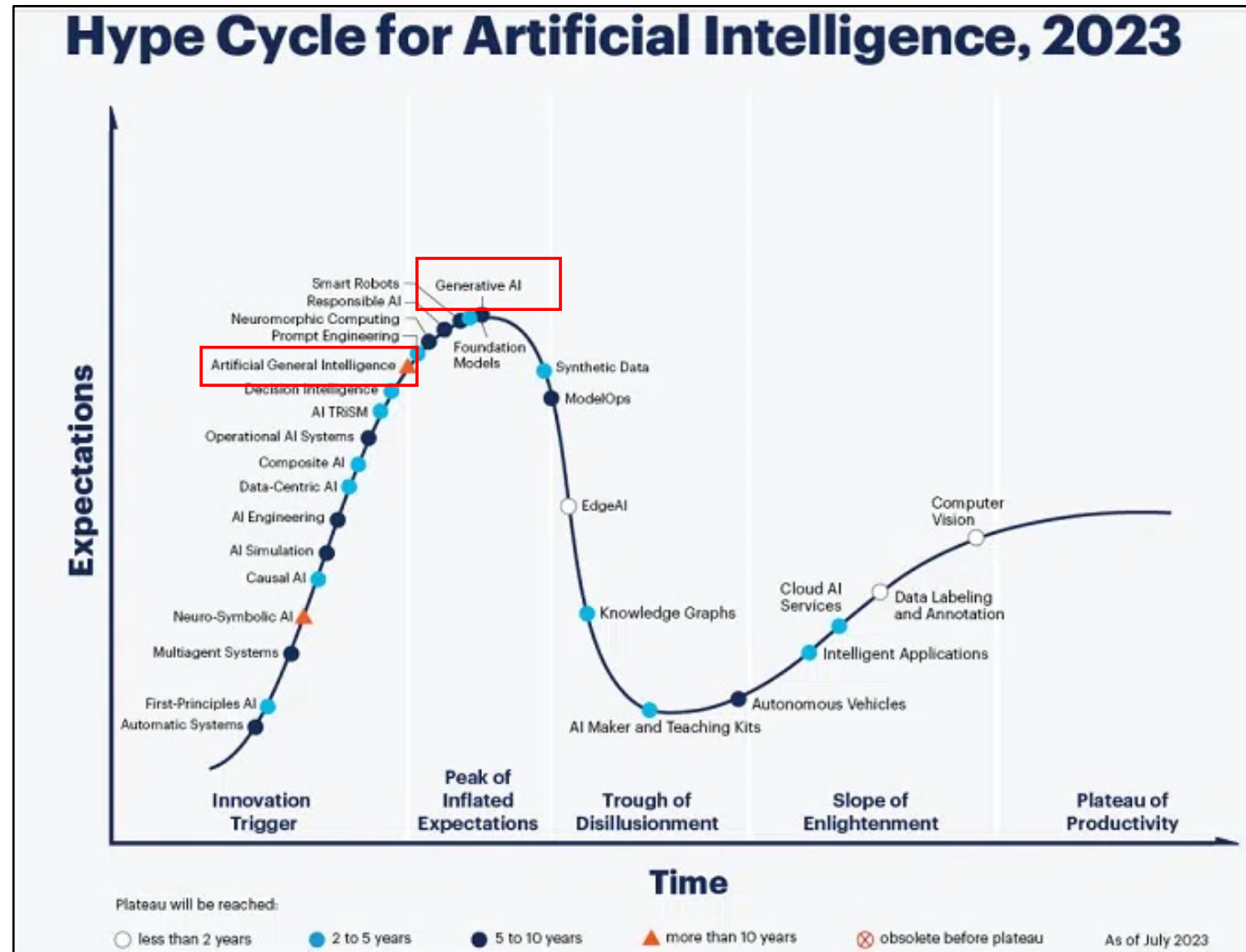
90ies

Today

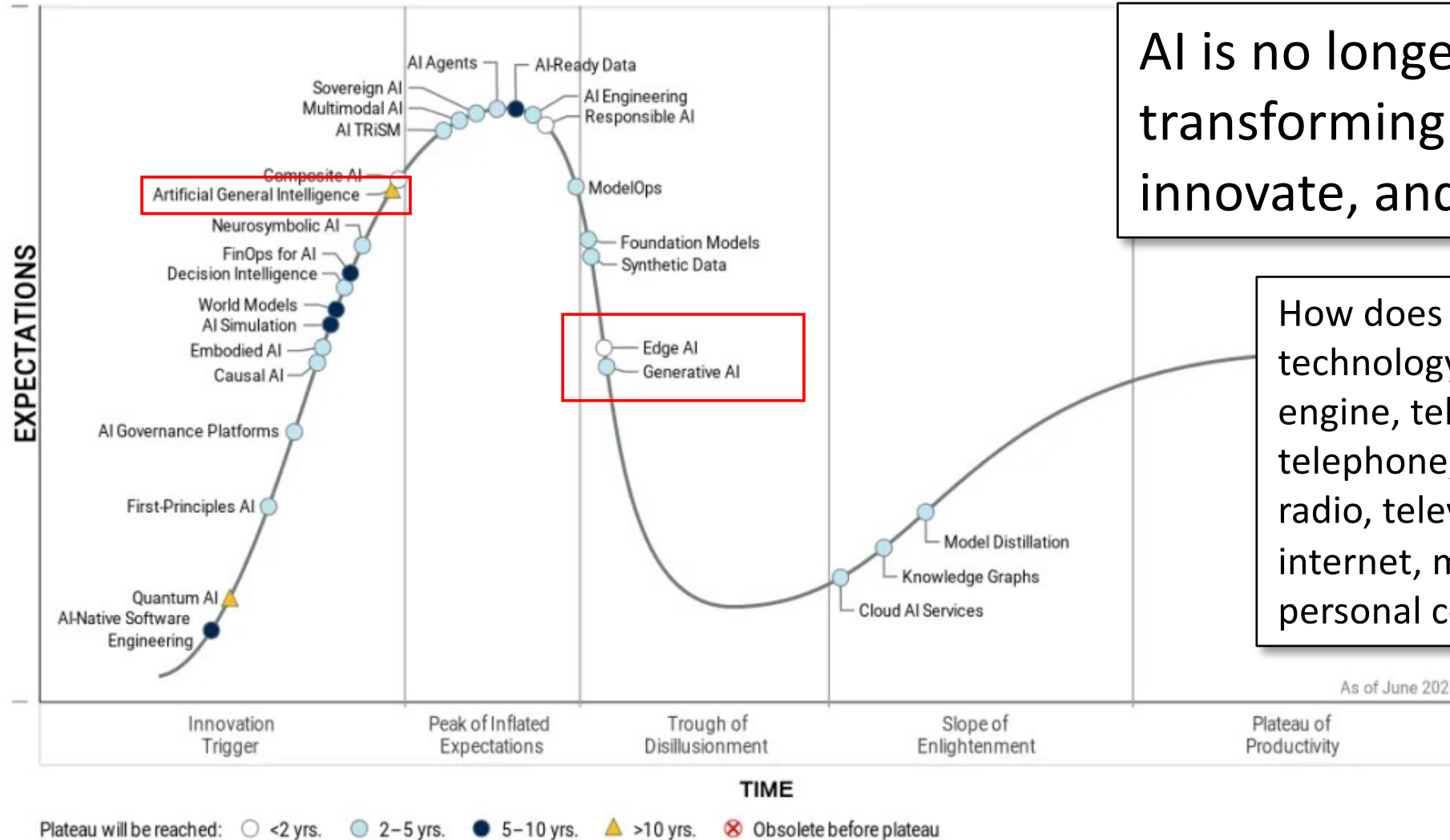
The future...

...disappearing AI





Hype Cycle for Artificial Intelligence, 2025



AI is no longer hype—it's transforming how we work, innovate, and invest.

How does AI as a revolutionary technology compare to the steam engine, telegraph, photograph, telephone, automobile, airplane, radio, television, computer, the internet, mobile phone, and the personal computer?

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NEWS IN BRIEF

Man Honestly Better Off For Having Turned Self Over To Algorithms

Published October 26, 2021





BOSTON—Noting the major improvements in his mood, taste, and overall outlook, friends and family members of local man Joseph Bennington told reporters Tuesday that he was honestly much better off for having turned his life over to the internet's algorithms. “He’s always in a great mood from looking at funny videos, and those sweaters that Amazon suggested he buy are a huge upgrade on his old work outfits,” said close friend Andy Wilcolm, explaining that Bennington’s life had improved precipitously over the past year as he exchanged free will for the automated decisions handed to him by algorithms at TikTok, YouTube, Netflix, Facebook, Spotify, Google, and other major services. “He also has a constant influx of pornography, tailored exactly to his taste, which means he doesn’t really have to go out and date. That had just been making him sad. But even if he did want to date, the algorithms at Tinder and Hinge would have him covered.” Sources close to the man added that they had also seen a marked improvement in his social life after the algorithms helped him find community with the Proud Boys, 8_Ch, and the Ku Klux Klan.



Christof Teuscher



teuscher@pdx.edu

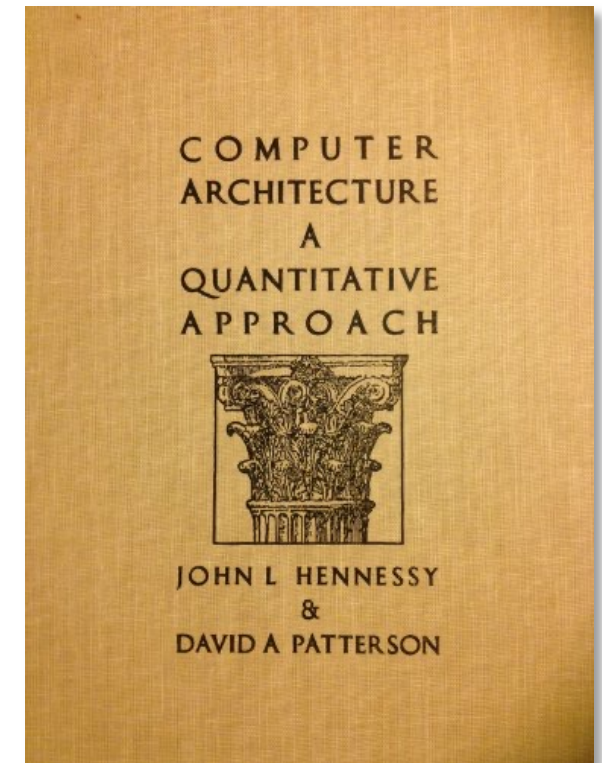
www.teuscher-lab.com/teaching



Traditional computer engineering and architecture is dead

AI/ML moves fast

- “In AI, something breaks, ships, or goes viral every six hours.”
- Timescale: ~weeks
- For comparison: Hennessy & Patterson, first edition, 1990
- I have a beautiful 2nd edition from 1996 that I bought for my computer architecture class.



Synopsys lays out strategy for AI 'agents' to design computer chips

By Stephen Nellis

March 19, 2025 10:30 AM PDT · Updated 10 days ago



"[...] shift from designing single chips to AI server systems with hundreds or even thousands of chips in them while aiming to release a new server each year"

"These are very complex and difficult to design," Ghazi said of new AI computers. "The pressure engineers are feeling today is not only complexity, **it is complexity and the pace by when they need to deliver these products, as well as the cost.**"

AgentEngineer: an "agent" that a human engineer can give instructions to.

<https://www.reuters.com/technology/artificial-intelligence/synopsys-lays-out-strategy-ai-agents-design-computer-chips-2025-03-19/>



AI-Driven Chip Design Inflection Point

Most advanced chips are now designed with AI-assisted tools

Agentic AI increases number of chips and further reducing time to market

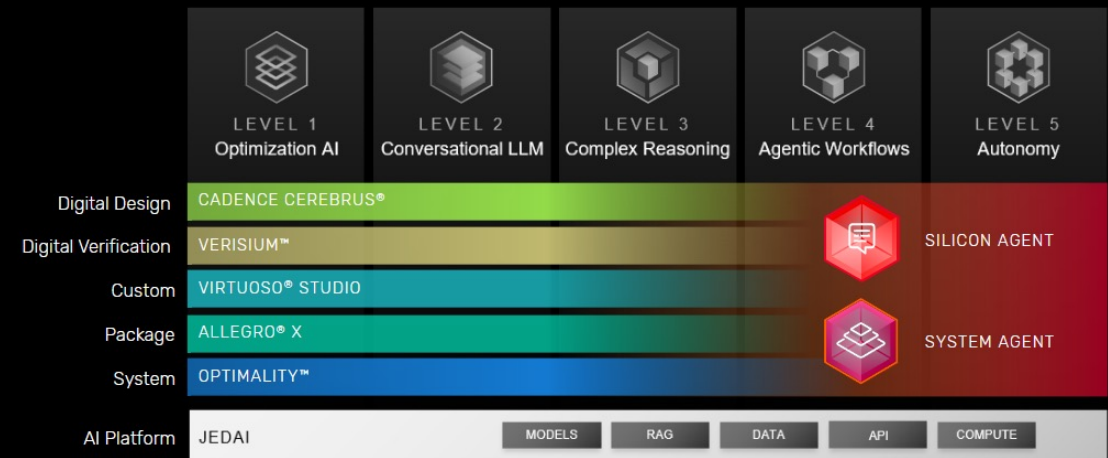
Agentic AI Accelerated

Chip Design Count (per year)

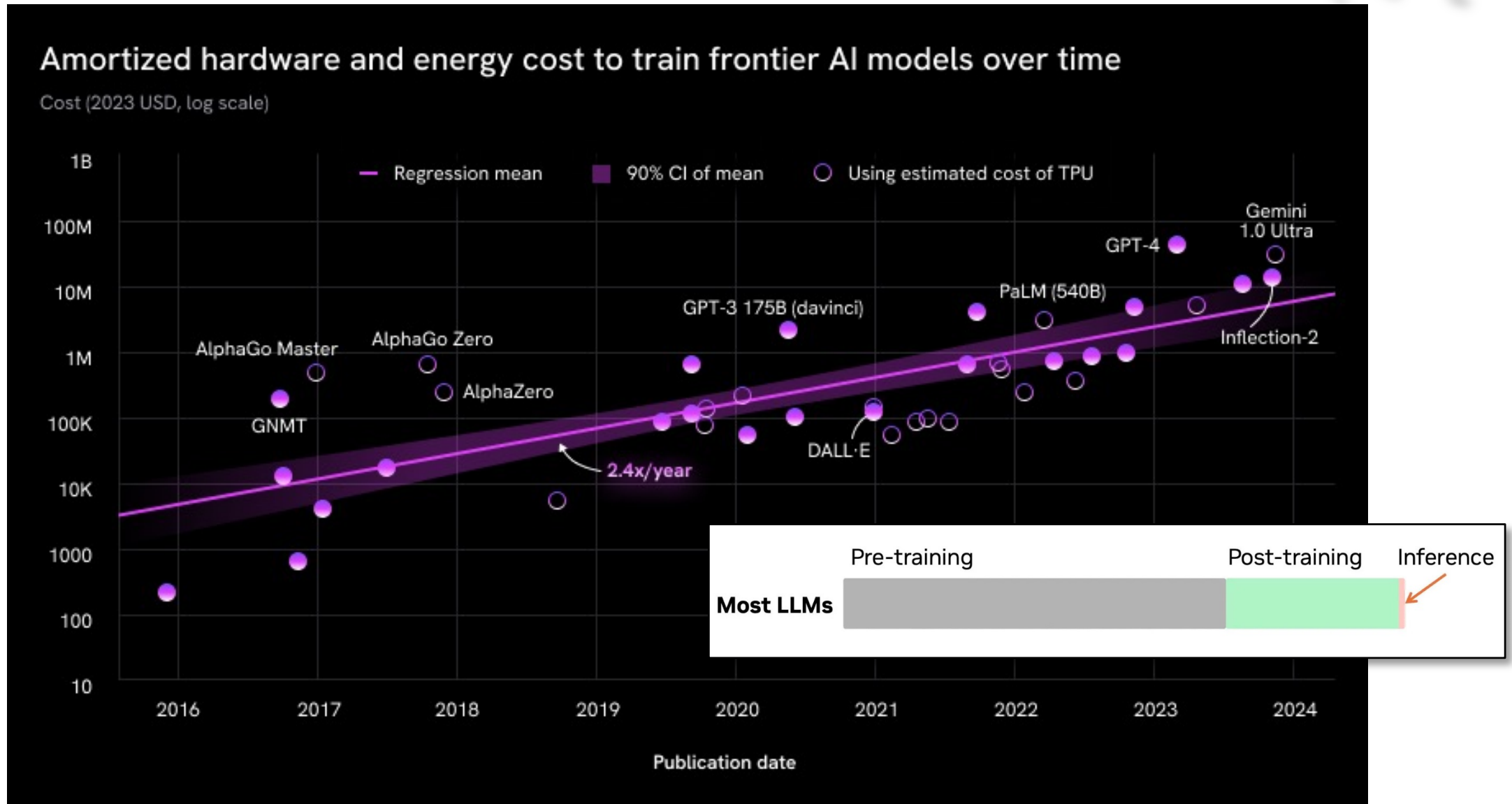
2020 2021 2022 2023 2024 2025e 2026e 2027e 2028e

Sources: Industry projections for tape-outs and AI adoption, and Cadence analysis.

The Journey to Autonomous Design



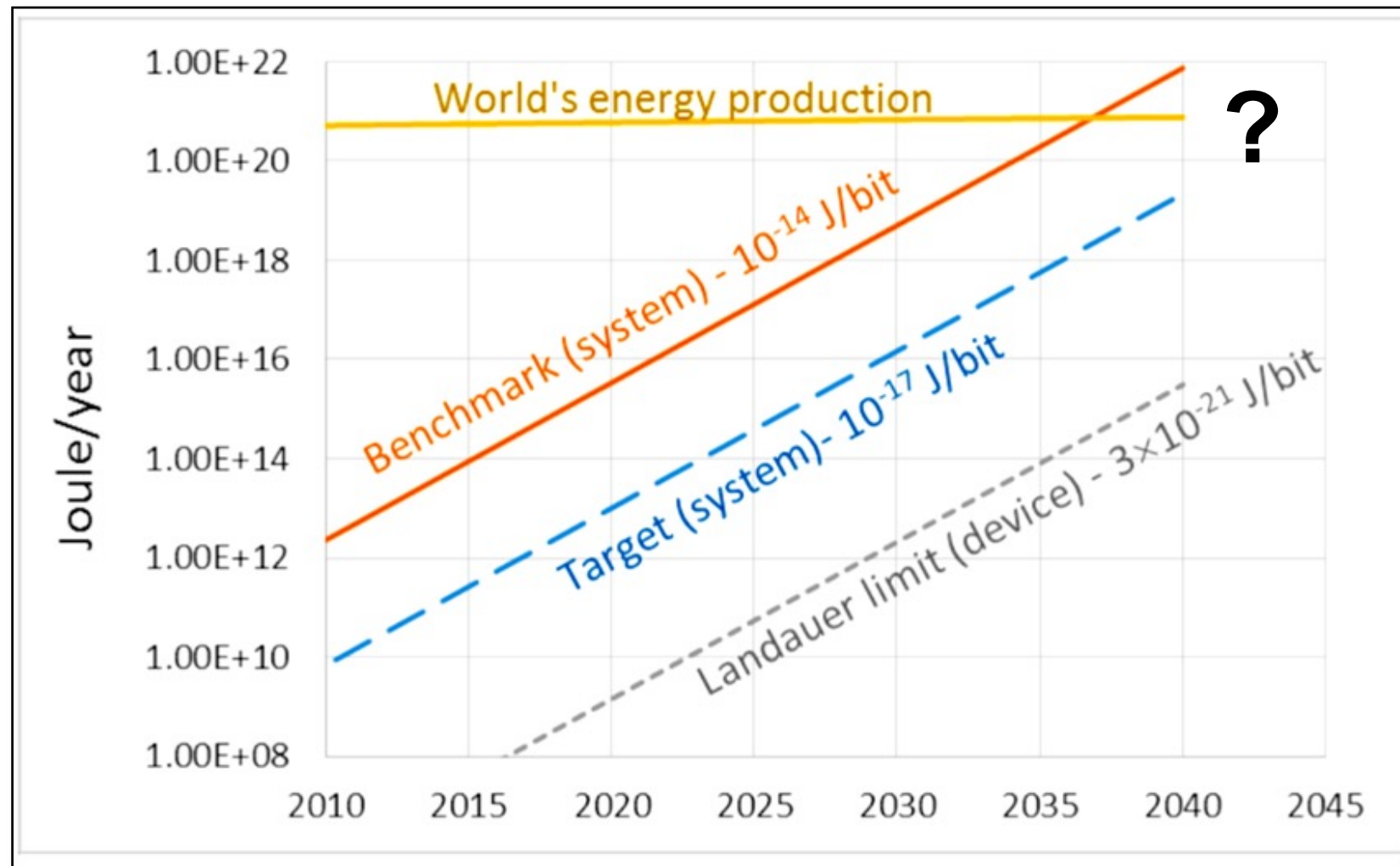
https://community.cadence.com/cadence_blogs_8/b/corporate-news/posts/agentic-ai-soc-system-engineering





What challenges do computer engineers face?

- A full stack understanding (algorithm to transistors)
- Co-design
- Deal with unseen complexities (server-scale AI machines, chiplets, trillions of transistors,...)
- Application-specific architectures for entirely new workloads
- Power barriers
- Shorter time to market
- Drastically increased need to keep up with state-of-the-art technological developments across the entire compute stack.
- Ethical issues
- ...



Estimated total energy expenditure for computing, directly related to the number of raw bit transitions. Source: SIA/SRC.

What skills do computer engineers have to have?

- **ML foundations.** A hardware engineer who cannot reason about tensor operations, quantization, sparsity, and memory access patterns cannot make good architectural decisions for ML **workloads**. Chip designers who use AI/ML as a black box will not be able to judge whether its outputs are reasonable or correct.
- **Hardware-software co-design.** This includes understanding how compilers, runtimes, and frameworks like TensorFlow or PyTorch map onto hardware primitives. The engineer needs to trace a **workload** from algorithm to register file.
- **Energy-aware design.** This spans from device-level techniques (voltage scaling, clock gating) through micro-architecture through system-level power management, with awareness of how **workload** characteristics interact with power budgets.
- **AI-assisted EDA.** Generative AI is now embedded in synthesis, verification, and physical design tools. Using these tools effectively requires enough domain knowledge to prompt them well and to audit their outputs. The shift from "run the tool" to "evaluate what the tool produced" is significant.
- **Verification at scale.** As SoC complexity grows and multi-die chiplet assemblies become common, verification is one of the hardest open problems. Engineers need both formal and simulation-based methods, and increasingly AI-assisted coverage analysis.

Foundation vs applied

- By the end of this term, most tools you will have used will already have changed and/or will be obsolete.
- **What stays is the foundations.**
- Knowing state-of-the-art industry tools is NOT a career strategy.
- Being successful on the job market means you can adapt to new tools because you know the foundations.



Progress in AI/ML is currently largely dependent on advances in hardware

Thus this course.

That's why this is a timely topic of interest to any computer engineer.

Co-design



INSPIRATION/MOTIVATION

CO-DESIGN APPROACH ON NEW
ENERGY-EFFICIENT MICROELECTRONICS

RESEARCH THRUST AREAS



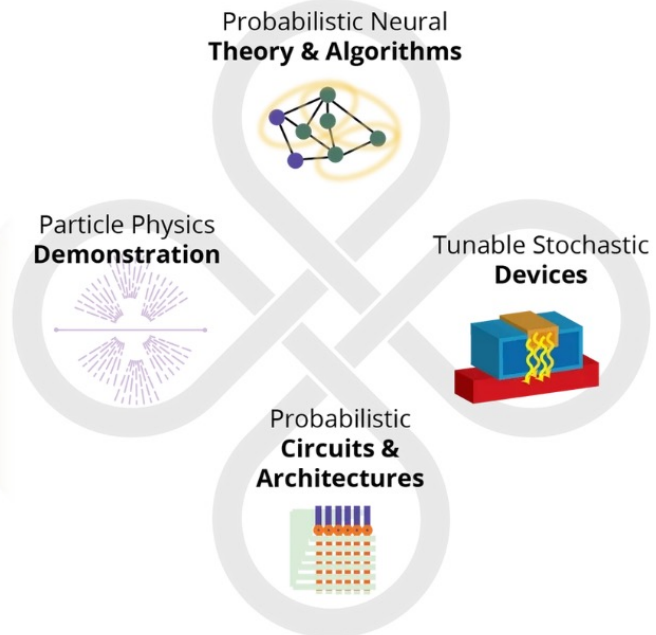
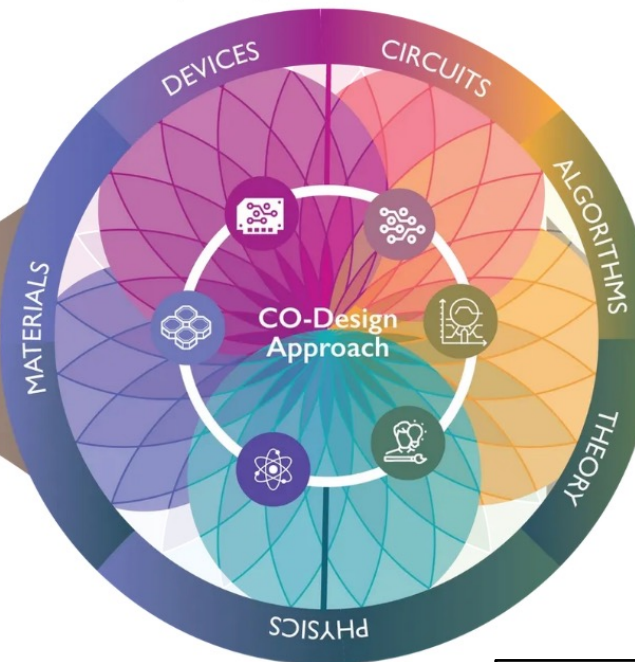
COINFLIPS

We have deterministic computing covered...We need probabilistic computing technologies



Every synapse in the brain is a stochastic "coinflip"

Co-design is proving invaluable in developing a novel paradigm for microelectronics



<https://coinflipscomputing.org>